

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 04 · Binomial Expansion

Questions

Student book · questions with writing space

- 41 pages
- Past papers up to January 2026

PURE 2 | TOPIC 04

Binomial Expansion

Your classified route

19

QUESTIONS

124

MARKS

3

FAMILIES

3

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 04 | FAMILY 01

Approximation

WHAT THIS FAMILY ASKS

Recognise the approximation demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

1 classified questions

8 total marks

4. (a) Find, in ascending powers of x , up to and including the term in x^3 , the binomial expansion of

$$\left(2 + \frac{x}{8}\right)^{13}$$

fully simplifying each coefficient.

(4)

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(b) Use the answer to part (a) to find an approximation for 2.0125^{13}

Give your answer to 3 decimal places.

(3)

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Without calculating 2.0125^{13}

(c) state, with a reason, whether the answer to part (b) is an overestimate or an underestimate.

(1)

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TOPIC 04 | FAMILY 02

Coefficients

WHAT THIS FAMILY ASKS

Recognise the coefficients demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**8 classified questions**

50 total marks

4. (a) Find the first three terms, in ascending powers of x , of the binomial expansion of

$$(2 + px)^6$$

where p is a constant. Give each term in simplest form.

(4)

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Given that in the expansion of

$$\left(3 - \frac{1}{2}x\right)(2 + px)^6$$

the coefficient of x^2 is $-\frac{3}{4}$

(b) find the possible values of p .

(4)

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1. The first three terms, in ascending powers of x , of the binomial expansion of $(1 + kx)^{16}$ are

$$1, -4x \text{ and } px^2$$

where k and p are constants.

- (a) Find, in simplest form,

(i) the value of k

(ii) the value of p

(3)

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DR ESAM AHMED

1. The first three terms, in ascending powers of x , of the binomial expansion of $(1 + kx)^{16}$ are

$$1, -4x \text{ and } px^2$$

where k and p are constants.

$$g(x) = \left(2 + \frac{16}{x}\right)(1 + kx)^{16}$$

Using the value of k found in part (a),

- (b) find the term in x^2 in the expansion of $g(x)$.

(3)

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3. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$\left(2 - \frac{kx}{4}\right)^8$$

where k is a non-zero constant. Give each term in simplest form.

(4)

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(3)

2. A curve C has equation $y = f(x)$ where

$$f(x) = (2 - kx)^5$$

and k is a constant.

Given that when $f(x)$ is divided by $(4x - 5)$ the remainder is $\frac{243}{32}$

- (a) show that $k = \frac{2}{5}$

(2)

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DR ESAM AHMED

2. A curve C has equation $y = f(x)$ where

$$f(x) = (2 - kx)^5$$

and k is a constant.

Given that when $f(x)$ is divided by $(4x - 5)$ the remainder is $\frac{243}{32}$

(b) Find the first three terms, in ascending powers of x , of the binomial expansion of

$$\left(2 - \frac{2}{5}x\right)^5$$

giving each term in simplest form.

(3)

Adaptive unit 2 · WMA12-01_2022_Oct_Q02

DR ESAM AHMED

2. A curve C has equation $y = f(x)$ where

$$f(x) = (2 - kx)^5$$

and k is a constant.

Given that when $f(x)$ is divided by $(4x - 5)$ the remainder is $\frac{243}{32}$

Using the solution to part (b) and making your method clear,

(c) find the gradient of C at the point where $x = 0$

(2)

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DR ESAM AHMED

4. The binomial expansion, in ascending powers of x , of

$$(3 + px)^5$$

where p is a constant, can be written in the form

$$A + Bx + Cx^2 + Dx^3 \dots$$

where A , B , C and D are constants.

- (a) Find the value of A

(1)

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DR ESAM AHMED

4. The binomial expansion, in ascending powers of x , of

$$(3 + px)^5$$

where p is a constant, can be written in the form

$$A + Bx + Cx^2 + Dx^3 \dots$$

where A, B, C and D are constants.

Given that

- $B = 18D$
- $p < 0$

(b) find

(i) the value of p

(ii) the value of C

(6)

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2. Find the coefficient of the term in x^7 of the binomial expansion of

$$\left(\frac{3}{8} + 4x\right)^{12}$$

giving your answer in simplest form.

(3)

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2. The first 4 terms, in ascending powers of x , in the binomial expansion of

$$(1 + px)^{10}$$

are

$$1 + 15x + qx^2 + rx^3$$

where p , q and r are constants.

Find the value of p , the value of q and the value of r .

(6)

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1. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(3 + kx)^7$$

where k is a non-zero constant. Give each term in simplest form.

(4)

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Given that, in this expansion, the coefficient of x^2 is 3 times the coefficient of x^3
(b) find, in simplest form, the value of k .

(2)

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TOPIC 04 | FAMILY 03

Expansion Terms

WHAT THIS FAMILY ASKS

Recognise the expansion terms demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

10 classified questions

66 total marks

- $$\left(1 + \frac{x}{4}\right)^{12}$$

(3)

(b) Find the term independent of x in the expansion of

$$\left(\frac{x^2 + 8}{x^5}\right)\left(1 + \frac{x}{4}\right)^{12}$$

(3)

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- (4)

2. One of the terms in the binomial expansion of $(3 + ax)^6$, where a is a constant, is $540x^4$

(b) Hence find the term independent of x in the expansion of

$$\left(\frac{1}{81} + \frac{1}{x^6}\right)(3 + ax)^6$$

(3)

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DR ESAM AHMED

1. Find the first four terms, in ascending powers of x , of the binomial expansion of

$$\left(2 + \frac{3}{8}x\right)^{10}$$

Give each coefficient as an integer.

(4)

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DR ESAM AHMED

3.

$f(x) = \left(2 + \frac{kx}{8}\right)^n$ where k is a non-zero constant

- (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of $f(x)$.
Give each term in simplest form.

(4)

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DR ESAM AHMED

3. $f(x) = \left(2 + \frac{kx}{8}\right)^n$ where k is a non-zero constant

Given that, in the binomial expansion of $f(x)$, the coefficients of x , x^2 and x^3 are the first 3 terms of an arithmetic progression,

(b) find, using algebra, the possible values of k .

(Solutions relying entirely on calculator technology are not acceptable.)

(3)

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10. (i) (a) Find, in ascending powers of x , the 2nd, 3rd and 5th terms of the binomial expansion of

$$(3 + 2x)^6$$

(3)

For a particular value of x , these three terms form consecutive terms in a geometric series.

(b) Find this value of x .

(3)

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(ii) In a **different** geometric series,

- the first term is $\sin^2 \theta$
- the common ratio is $2 \cos \theta$
- the sum to infinity is $\frac{8}{5}$

(a) Show that

$$5 \cos^2 \theta - 16 \cos \theta + 3 = 0 \quad (3)$$

(b) Hence find the exact value of the 2nd term in the series.

(3)

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1. (a) Find the first four terms, in ascending powers of x , of the binomial expansion of

$$\left(1 - \frac{1}{6}x\right)^9$$

giving each term in simplest form.

(3)

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(b) Hence find the coefficient of x^3 in the expansion of

$$(10x + 3)\left(1 - \frac{1}{6}x\right)^9$$

giving the answer in simplest form.

(2)

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4. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(3 + 2x)^6$$

giving each coefficient in simplest form.

(4)

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(b) Hence find the coefficient of x^2 in the expansion of

$$\left(2x^2 - \frac{1}{6x}\right)(3 + 2x)^6$$

(3)

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5. (a) Find, in terms of a , the first 3 terms, in ascending powers of x , of the binomial expansion of

$$(2 + ax)^6$$

where a is a non-zero constant. Give each term in simplest form.

(3)

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DR ESAM AHMED

$$f(x) = \left(3 + \frac{1}{x}\right)^2 (2 + ax)^6$$

Given that the constant term in the expansion of $f(x)$ is 576

(b) find the value of a .

(4)

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2. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of $(2 - 5x)^8$ giving each term in simplest form.

(4)

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This expansion is to be used to find an approximation for 2.05^8

(b) State the value of x that should be used.

(There is no need to carry out this calculation.)

(1)

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1: (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(1 - 4x)^7$$

giving each term in simplest form.

(4)

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In the series expansion of $(5 + kx)(1 - 4x)^7$ where k is a constant
the coefficient of the term in x^2 is 1316
(b) Use the answer to part (a) to find the value of k .

(2)

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